

Mueez Mughal | Mechatronics Engineer

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Education

University of British Columbia, Vancouver, BC

Expected May 2028

Bachelor of Applied Science - Mechanical Engineering (Mechatronics Specialization)

Dean's List

UBC Applied Science Co-op Program

Skills

Software: MATLAB, Python, C, Simulink, SolidWorks, ANSYS, KiCad, LTSpice, LabVIEW, Git, 8051 Assembly

Fabrication: CNC Milling, Lathe, SMD/THT Soldering, 3D Printing (FDM/SLA), Composite Fabrication

Mechanical: Mechanical Design, FEA, GD&T, Tolerance Stack-Up, Structural & Fatigue Analysis, Machine Design

Electronic: STM32, ESP32, Arduino, Circuit Design, PCB Design, Motor Control, Power Electronics, VHDL, Verilog, FPGA

Communication: Technical Writing, Cross-Functional Collaboration, Documentation, Leadership

Work Experience

Fingerprint Technologies, Vancouver, BC

May 2026 – Aug 2026

Mechatronics R&D Engineering Intern

- Designed a mixed-signal control PCB for an electromechanical system, from schematic through fabrication-ready release.
- Integrated analog sensing with power electronics and stepper motor stages, applying noise-aware layout and grounding.
- Designing mechanical test fixtures and validation rigs to evaluate chassis performance against European standards.
- Performed end-to-end mechanical design for prototype and production assemblies, spanning chassis and machine levels.
- Applied Lamé's equations to estimate press-fit forces, then validated theoretical behavior against measured loads.

Suncor Energy, Calgary, AB

May 2025 – Dec 2025

Mechanical Engineering Intern

- Developed a physics-based Python diagnostic platform for ESP induction motors, enabling real-time snapped-shaft detection.
 - Modeled downhole-to-wellhead mass and energy balances to reconcile electrical input power with mechanical shaft output.
 - Analyzed motor current signatures, power factor, and reactive power flow to diagnose electromechanical failure modes.
 - Built induction motor equivalent circuit models to evaluate actuator torque and efficiency under varying load conditions.
 - Automated ESP lifecycle performance analytics in Python, contributing to \$2.5M in sustained annual savings.
 - Created real-time PI Vision dashboards from P&IDs to accelerate fault isolation, validation, and operational decision-making.
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Extracurriculars

UBC Solar Racing, Vancouver, BC

6th Place Overall (6/24) — Qualified for American Solar Challenge

Vehicle Dynamics Lead

Sep 2025 – Present

- Lead a 13-member Vehicle Dynamics team owning suspension, steering, and braking for UBC Solar's Gen 4 car.
- Set technical direction across CAD, FEA, testing, validation, and manufacturing for vehicle dynamics systems.
- Review analyses, CAD, and drawings to ensure performance, safety, and manufacturability requirements are met.
- Drive cross-functional design reviews to resolve integration issues and deliver systems ahead of design freeze.

Vehicle Dynamics Sub-Team Member

Sep 2024 – Sep 2025

Steering Wheel Design

- Designed and integrated a competition-compliant quick-release steering wheel system, reducing driver egress time by 30% while preserving steering stiffness, structural integrity, and all safety requirements under race conditions.
- Collaborated closely with the Power and Signals team to embed, route, and mechanically integrate a custom PCB within the steering assembly, improving signal reliability, robustness, and overall driver interface quality.
- Fabricated aluminum baseplates using waterjet cutting and produced multiple rapid prototypes via 3D printing to validate fit, ergonomics, tolerance stack-up, and final assembly repeatability.

Lower Control Arm Design (Gen 4 Car)

- Conducted R&D on the lower control arms for UBC Solar's 4th-generation vehicle.
 - Performed ANSYS FEA under 2g bump, 1g cornering, and 1g braking load cases to validate static and fatigue strength.
 - Conducted destructive testing on suspension rod ends to determine yield strength and characterize failure modes.
 - Produced manufacturing drawings with GD&T to control tolerance stack-up and ensure repeatable fabrication.
 - Developed a quarter-car suspension model in MATLAB using Laplace-domain analysis to identify critical ride frequencies, assess system dynamics, and validate spring-damper selection against performance targets.
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Projects

Homemade Capacitive Weight Sensor

Nov 2024

- Developed a first-principles model linking capacitance ($C = \kappa A/d$) to applied load through solid mechanics analysis of polyurethane foam deformation.
- Designed and built a parallel-plate capacitor sensor capable of measuring applied weight with ~99% accuracy.